MANO HARSHA DAVULURI

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PROFESSIONAL SUMMARY

Results-driven **NITian** and **AI & Data Engineer** with 3 years of experience building production Generative AI applications and scalable data platforms. Specializes in agentic AI (MCP, LLM tool-calling, RAG with vector search), LLMOps with MLflow, and FastAPI services on Databricks, with a strong foundation in Spark, AWS, data lakes, and ETL pipelines. Architected and delivered an enterprise conversational AI assistant that enables natural-language data discovery across data assets, products, and domains. Proven track record of leading cross-functional teams and driving AI-led innovation with data governance at enterprise scale.

TECHNICAL SKILLS

Generative AI & Agents: Agentic AI, RAG, MCP (FastMCP), Agent-to-Agent, LangChain, LangGraph, ChatDatabricks, Prompt Engineering, Guardrails, Embeddings, Hybrid Search (pgvector HNSW, RRF).
LLMOps & AI Platforms: MLflow 3 Tracing & LLM Judges, Databricks Model Serving, AI Gateway, Vector Search, Databricks Apps, AI Index.
Cloud & Platforms: Databricks (Unity Catalog, Lakebase, Delta Lake, SQL Warehouses, Workflows), AWS (S3, Lambda, EKS).
Data Engineering: Apache Spark, PySpark, Spark SQL, Delta Lake (CDF, MERGE), Brickflow, Koheesio, SparkExpectations, DBT, Informatica CDGC, PostgreSQL.
Data Modeling: Dimensional Modeling, Star/Snowflake Schema, SCD Type 1/2, Medallion, Lakehouse, Data Governance.
Programming & DevOps: Python, SQL, FastAPI, Pydantic, Git, Docker, Jenkins CI/CD, OAuth 2.0 (Okta, M2M), Cerberus, pytest.

PROFESSIONAL EXPERIENCE

Nike | *Software Engineer II – AI* *Bengaluru, India | Jan 2026 – Present*

Compass AI – Agentic chatbot for Nike's enterprise data catalog

- Built and extended an **MCP (Model Context Protocol)** server for a **LangChain AI agent** on Databricks Apps to **12 LLM-callable tools**, incl. an **agent-to-agent** tool calling Nike's AirTalk chatbot on the user's behalf via **Okta OAuth**, and tools for AI-proposed data products, docs lookup, and catalog stats.
- Engineered retrieval tools over **Lakebase (Postgres)** with **hybrid semantic search** (pgvector + full-text via **Reciprocal Rank Fusion**) and safe SQL access; replaced an external REST API with direct table queries, removing a network hop and its auth.
- Implemented end-to-end **MLflow 3 tracing**, propagating request/session IDs from FastAPI to the MCP server via HTTP headers so every LLM and tool call in a multi-turn chat forms one traced session for production debugging.
- Established **LLM evaluation and monitoring** with MLflow 3 **LLM-as-a-judge** scorers (completeness, role adherence, tool-call efficiency, user frustration) on production sessions, streaming latency metrics (time-to-first-token), and **prompt caching** to cut token cost and latency.
- Engineered the agent's **system prompt and guardrails**: scope restriction to Nike metadata, non-disclosure of internals, correct bot identity, and accurate deep links to products, assets, and metrics.
- Productionized the MCP server across **dev/QA/prod** with **Jenkins CI/CD**, per-env config, **service-principal OAuth M2M**, and Cerberus secrets; retired a legacy MCP server and automated access-scope and vector index refresh.
- Built **FastAPI** chatbot features: chat-history APIs (sessions, timestamped recent queries, per-session limits), a live health endpoint for UI "Online" status, and resilient ServiceNow ticketing, at a **90% test coverage** gate.
- Delivered medallion **ETL pipelines** (Spark, Brickflow) and a gold-layer **dimensional model** (22 dims, 5 facts) with SparkExpectations DQ rules, grounding the agent across **100,000+** data assets.

Epsilon (Publicis Groupe) | *Data Engineer* *Bengaluru, India | Aug 2023 – Dec 2025*

- Designed and implemented comprehensive Data Quality framework for Prospect project, incorporating automated discrepancy detection and real time alerting mechanisms to ensure data integrity and compliance.
- Constructed scalable ETL/ELT data solutions processing over 150+ billion records daily into data lakes and warehouses using Spark, developed ML Models enhancing decision-making and resulting in 25% boost in quarterly revenue.
- Optimized and tuned Spark jobs for large-scale datasets (30+ TB), implementing performance best practices and reducing infrastructure costs by 46%.
- Collaborated with cross-functional teams to establish data governance standards and implement CI/CD pipelines for automated data processing workflows.
- Built end-to-end data pipelines encompassing Ingestion, extract-transform-load processes, data archiving, warehousing and reporting automation using Apache Spark on Databricks Delta tables.
- Collaborated with solution architects and business stakeholders to design and implement complex ML models for predictive analytics and BI solutions.

CERTIFICATIONS

- Databricks Certified AI Engineer Associate – Databricks (Issued 2026)
- Databricks Certified Data Engineer Associate – Databricks (Issued 2023)

EDUCATION

National Institute of Technology (NIT) Karnataka | *Aug 2019 - May 2023*
Bachelor of Technology | Surathkal, Karnataka